SHARE-ViT: Shared Hexagonal Angle–Scale Routing Enhancement for Vision Transformers

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Abstract

Vision Transformers typically begin with a square, axis-aligned patch projection. This front end has no explicit representation of local pose or scale, while absolute positional embeddings encode where a token is but not which image-conditioned direction it should attend to. We introduce SHARE-ViT, a geometric enhancement that couples a shared hexagonal angle–scale tokenizer with pose-conditioned attention bias. Images are sampled on a staggered hexagonal lattice, compared with learned variable-resolution polar prototypes at two scales and several orientations, and projected into compact cosine/sine pose coordinates. A null-softmax route allows a prototype to abstain when no tested pose is suitable. The same geometric language drives two directed attention mechanisms: Image Look estimates pose from the input image, whereas Feature Look estimates it directly from the paired tokenizer coordinates and can share one pose probe across several Transformer layers. In controlled 20-epoch training from scratch on ImageNet-100, the six-direction tokenizer with Image Look reaches 55.04% top-1 and 81.80% top-5 accuracy, compared with 51.52% and 79.12% for a standard DeiT-Tiny-sized baseline. Feature Look alone with PE reaches 55.18% top-1; combining both Look branches with a three-layer probe-sharing span reaches 55.54% top-1 and 81.82% top-5. These variants remain below the baseline parameter count. The results support explicit local pose and scale as a useful interface between image sampling and global attention, but do not establish exact equivariance or broad cross-dataset superiority.

1 Introduction

Vision Transformers (ViTs) apply self-attention to a sequence of image tokens [Dosovitskiy et al., 2021, Touvron et al., 2021]. Their global interaction is attractive, but the sequence is normally created by a square patch projection with one learned weight for every Cartesian location. On a square lattice, axial neighbors are one grid step away while diagonal neighbors are $\sqrt{2}$ steps away. Equal changes in discrete direction therefore do not correspond to equal spatial displacements. The tokenizer also entangles local appearance with one canonical orientation and scale.

Recent tokenization studies make this front-end choice increasingly explicit. FlexiViT interpolates patch-embedding parameters across patch sizes [Beyer et al., 2023], while MSViT selects mixed token scales according to image content [Havtorn et al., 2023]. These methods improve scale flexibility, but they do not explicitly represent the orientation of a matched local pattern.

Position encoding addresses a related but different issue. Absolute embeddings tell the model where a token occurs, and relative position methods tell attention how two token indices are displaced [Wu et al., 2021]. More recent work extends rotary embeddings to two-dimensional ViT coordinates [Heo et al., 2024] and redesigns tokenization, attention, patch merging, and positional encoding for exact circular shift equivariance [Rojas-Gomez et al., 2024]. Neither mechanism, by itself, asks whether the image evidence supports a particular pose, nor does it convert that evidence into a directed search pattern. Local-window Transformers such as Swin introduce efficient spatial locality [Liu et al., 2021], whereas conditional position encodings generate input-dependent position signals [Chu et al., 2021]. Our goal is complementary: construct an explicit local angle–scale representation before global attention and reuse it to condition where each attention head looks.

Rotation-aware convolution has a long history. Group-equivariant networks share filters under discrete transformations [Cohen and Welling, 2016], active rotating filters materialize oriented responses [Zhou et al., 2017], and harmonic networks encode rotations with circular harmonics [Worrall et al., 2017]. Hexagonal convolution reduces lattice anisotropy and permits additional rotational weight sharing [Hoogeboom et al., 2018]. Recent Transformer designs carry this program further through discrete $E(2)$-equivariant attention [Xu et al., 2023], continuous harmonic representations [Karella et al., 2024], and steerable Fourier features [Kundu and Kondor, 2025]. SHARE-ViT builds rotation-shared local matching on a Hex lattice and then transfers the resulting pose evidence to Transformer attention. It is a pose-aware enhancement of a ViT rather than a strict group-equivariant model.

Our contributions are:

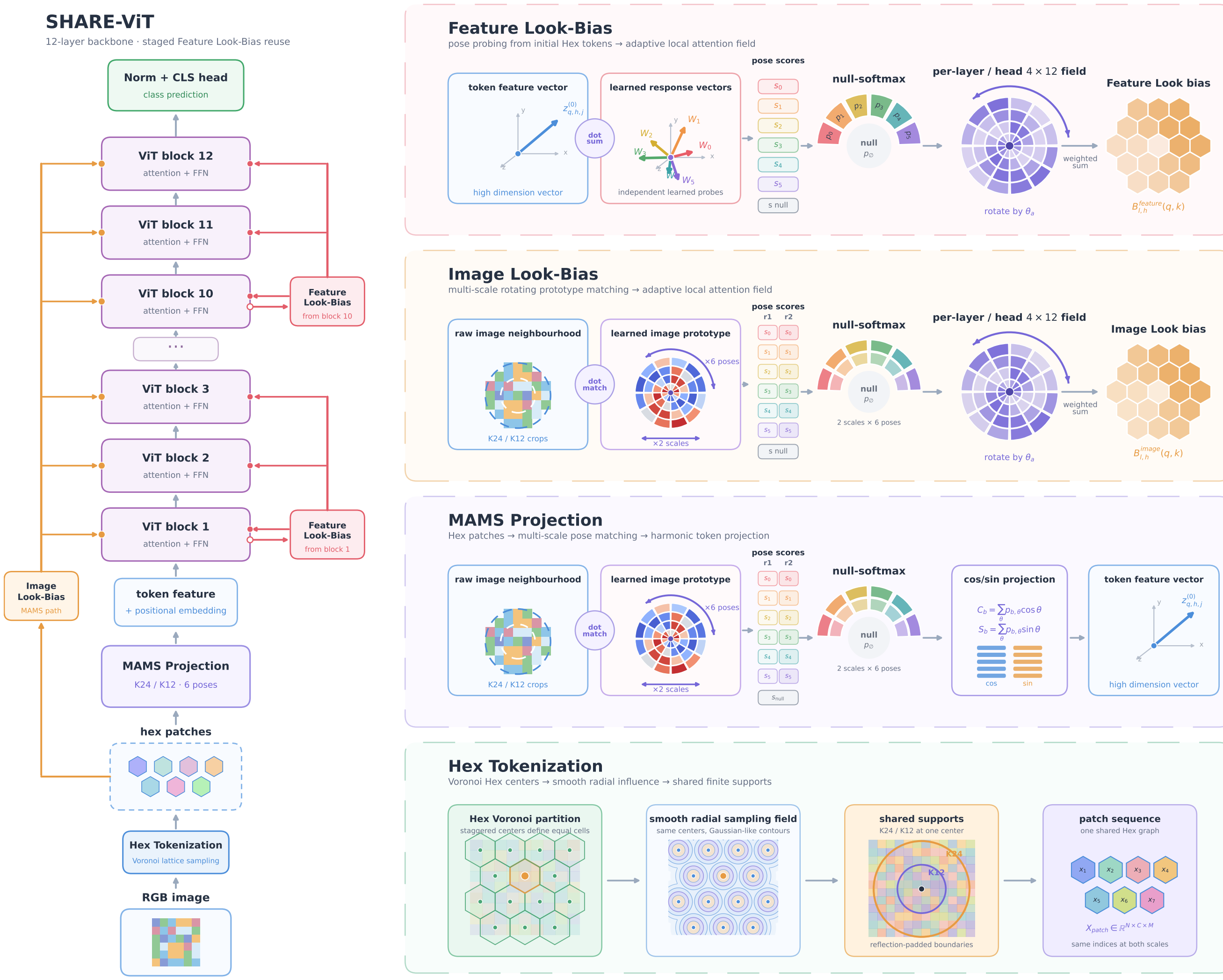

Figure 1: Overview of SHARE-ViT. Hex Tokenization constructs the shared lattice, MAMS Projection extracts multi-scale multi-pose token features, and Image and Feature Look paths inject complementary directed attention biases across the Transformer backbone.

- a Hex-centered, multi-scale rotation-matching tokenizer: shared variable-resolution polar prototypes are evaluated over discrete poses, a null route suppresses unsupported matches, and circular moments preserve compact directional features on a common token graph;
- two pose-conditioned attention biases derived from the same geometric representation: Image Look reads direction and scale from raw-image patches, while Feature Look reads pose from tokenizer features and allows probe reuse across nearby Transformer layers. Thus local pose evidence is used both to construct tokens and to direct later global attention.

2 Related Work

Visual tokenization and spatial bias. ViT maps non-overlapping square patches to tokens and adds learned positional embeddings [Dosovitskiy et al., 2021]. DeiT demonstrates effective training of this architecture on ImageNet without external pretraining [Touvron et al., 2021]. Later models inject more local structure: Swin restricts attention to shifted windows [Liu et al., 2021], iRPE introduces directional relative position terms [Wu et al., 2021], and CPVT generates conditional positional encodings from local neighborhoods [Chu et al., 2021]. RoPE-ViT adapts rotary coordinates to two-dimensional visual tokens and improves resolution extrapolation [Heo et al., 2024], while recent shift-equivariant ViTs treat tokenization and positional encoding as part of the equivariance problem [Rojas-Gomez et al., 2024]. SHARE-ViT preserves global attention. Locality enters through geometric token formation and an input-conditioned directed field, rather than a fixed window or only an index-based lookup table.

Flexible visual tokenization. Patch size controls both the visual support of a token and the sequence length. FlexiViT randomizes patch size and interpolates patch and position parameters so one ViT can operate at several resolutions [Beyer et al., 2023]. MSViT instead uses a learned gate to construct mixed-scale token sequences [Havtorn et al., 2023]. These approaches vary token scale or allocation; our tokenizer keeps one shared Hex token graph and evaluates multiple support scales and orientations at each center before projecting a token.

Rotation-aware representations. G-CNNs formalize weight sharing over transformation groups [Cohen and Welling, 2016]. ORNs rotate active filters and retain orientation channels [Zhou et al., 2017], while harmonic networks use circular bases to encode rotation [Worrall et al., 2017]. At the Transformer boundary, Equi-ViT replaces patch embedding with radially symmetric Gaussian-mixture-ring convolutions [Chen et al., 2026]; GE-ViT instead lifts tokens to a joint spatial–orientation domain and preserves a discrete group axis through self-attention [Xu et al., 2023]; and Steerable Transformers carry Fourier irreducible representations through attention, normalization, and feed-forward layers [Kundu and Kondor, 2025]. Harmformer combines a harmonic convolutional stem with attention to obtain continuous roto-translation equivariance [Karella et al., 2024]. These methods motivate our shared prototypes and cosine/sine pose coordinates, but our current tokenizer learns no A/B orthogonal value basis: it directly takes a fixed first circular moment of the routed pose scores. The present model samples a finite set of orientations and does not enforce an equivariance identity; we therefore use *pose-aware* and *rotation-shared*, rather than *rotation-equivariant*, in our claims.

Hexagonal image processing. Hexagonal lattices provide six equidistant first-ring neighbors. HexaConv shows that hexagonal planar and group convolutions can reduce lattice anisotropy and improve rotational parameter sharing [Hoogeboom et al., 2018]. Our input remains a conventional Cartesian image. We place patch centers on staggered rows and sample circular supports by interpolation, requiring neither native hexagonal camera data nor a hexagonal storage format.

3 Method

3.1 Overview

SHARE-ViT has a tokenizer and two optional directed-bias paths (Fig. 1). The tokenizer maps an image to a sequence of 192-D tokens. Image Look independently re-samples the input image; Feature Look instead interprets the token dimensions as geometric pairs. Both produce pose probabilities that rotate learned radial–angular fields before the fields are added to attention logits. The paths share a geometric language—scale, direction, polar sampling, and null routing—but do not have to share learned prototype parameters.

3.2 Hexagonal patch centers

Let $x \in \mathbb{R}^{H\times W\times 3}$ be an input image. Patch centers are placed on alternating horizontally shifted rows, producing a staggered Hex lattice. At 224×224 resolution, the current geometry yields 195 image tokens, close to the 196 tokens of a standard 14×14 DeiT tokenizer. Reflection padding handles image boundaries. Two circular supports with diameters $K \in \{24, 12\}$ share every center, so scale comparison does not alter the token graph.

Each support uses a cosine-shaped radial cover $c_K(u)$. We preserve its accumulated response magnitude rather than normalizing it to unit mass. The smaller support is compensated by

$$\gamma_K = \frac{\sum_u c_{24}(u)}{\sum_u c_K(u)}, \tag{1}$$

removing a response difference caused only by the number of covered pixels.

3.3 Variable-resolution polar prototypes

The tokenizer contains $P = 96$ learned prototypes. Each prototype is stored in polar coordinates over $R = 12$ rings. We denote a half-turn geometry allocation by $\mathcal{G}_{D,q}$, where D is the number of tested poses and q is the base number of angular samples added per radius. Under $\mathcal{G}_{D,q}$, ring r stores $N_r = q(r + 1)$ samples and pose d uses $\theta_d = \pi d/D$, for $r = 0, \ldots, R - 1$ and $d = 0, \ldots, D - 1$. The tested poses therefore uniformly cover one half turn. Inner rings avoid angular parameters that cannot

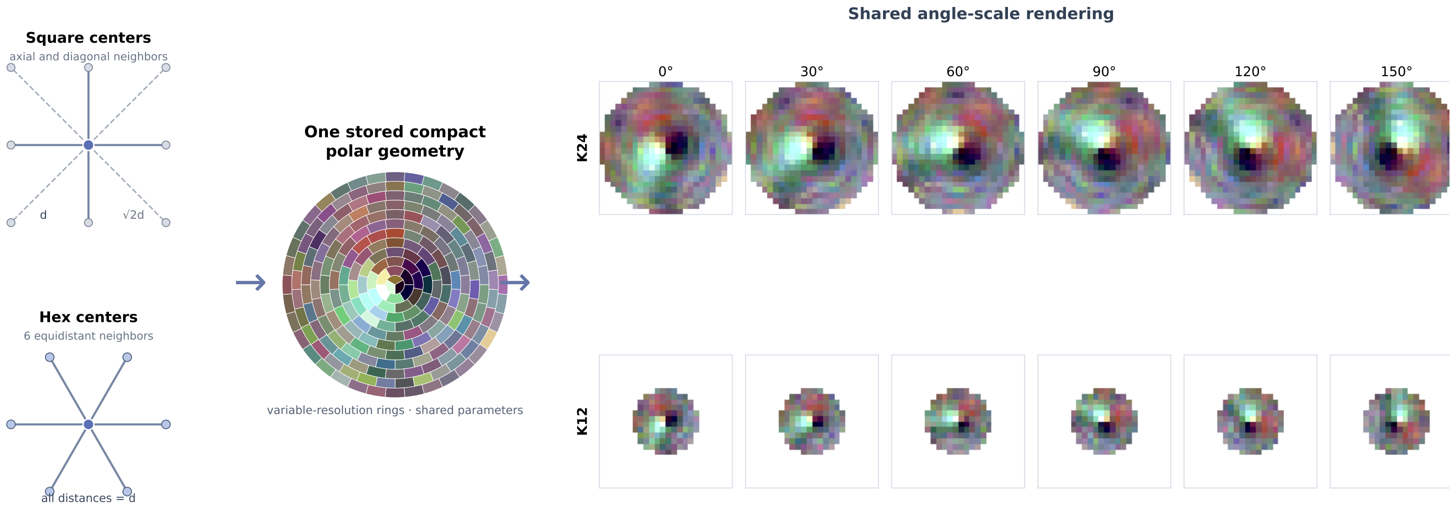

Figure 2: Hex-centered and shared angle–scale geometry. A compact variable-resolution polar prototype is rendered at two support sizes and six half-turn poses. The displayed kernels come from a trained model; the figure illustrates parameter sharing rather than an equivariance proof.

be spatially resolved, while outer rings retain more detail. Bilinear polar-to-Cartesian interpolation renders each prototype at a requested scale and pose (Fig. 2).

For token n, prototype p, scale k, and tested direction d, the raw matching score is the covered dot product

$$s_{npkd}^{\mathcal{G}_{D,q}} = \sum_{u \in \Omega_k} \gamma_k c_k(u) x_{n,k}(u)^\top w_{p,k,d}^{\mathcal{G}_{D,q}}(u). \tag{2}$$

No division by $\|x\|\|w\|$ is applied: this is a dot-product kernel, not cosine similarity. Scores from both supports are accumulated before routing, $s_{npd} = \sum_k s_{npkd}$.

The two evaluated allocations are $\mathcal{G}_{4,4}$, which tests four poses at 45° increments, and $\mathcal{G}_{6,3}$, which tests six poses at 30° increments. The corresponding full directional periods used by the Look fields contain $D_{\text{global}} = 2D$, hence eight and twelve bins. Direction count and stored ring density therefore change together; the current experiment compares these complete allocations and does not identify either factor in isolation.

3.4 Null-softmax circular-moment projection

A learned scalar $s_p^{\varnothing}$ is appended to the direction logits of each prototype. Let $\mathbf{p}_{np}$ contain the resulting pose probabilities and the null probability:

$$\mathbf{p}_{np} = \mathrm{softmax}([s_{np0}, \dots, s_{np,D-1}, s_p^{\varnothing}]). \tag{3}$$

The null branch contributes zero, and the remaining direction probabilities are not renormalized. A prototype can therefore lower its total activation when none of the tested poses is appropriate.

Using the angles defined by $\mathcal{G}_{D,q}$, each prototype emits its first circular moment,

$$z_{np} = \sum_{d=0}^{D-1} p_{npd} \begin{bmatrix} \cos\theta_d & \sin\theta_d \end{bmatrix}^\top. \tag{4}$$

Concatenating z_{np} over 96 prototypes gives a 192-D token. This shares the output basis across directions instead of learning a separate 192-D value vector for every pose.

3.5 Image-conditioned Look bias

Image Look uses one raw-image probe for each of the $12 \times 3 = 36$ layer–head pairs. Image-to-ring extraction does not record image gradients, whereas probe and field parameters remain learnable. A probe returns null-softmax pose weights π_{bqhkd} for batch item b, query token q, head h, scale k, and direction d.

Each head stores a signed canonical field $G_h \in \mathbb{R}^{R_L \times D_L}$. Here $R_L = 2|\mathcal{K}| = 4$, and $D_L = 2D$ follows the tokenizer period: eight bins for $\mathcal{G}_{4,4}$ and twelve for $\mathcal{G}_{6,3}$. Rotating and radially interpolating G_h yields a field F_{hkdqj} over key token j. The attention bias is

$$B^{\text{img}}_{bhqj} = \sum_{k,d} \pi_{bqhkd} F_{hkdqj}. \tag{5}$$

The null route contributes zero. Since the field is conditioned on the query token's local evidence, generally $B_{bhqj} \neq B_{bhjq}$.

3.6 Token-conditioned Feature Look

Feature Look removes the raw-image probe and operates directly on the 96 cosine/sine pairs of each tokenizer output before adding positional embeddings. The tokenizer output bias is subtracted first. The pairs are divided evenly across three heads, giving 32 pairs per head. A probe group g learns six axis detectors A_{ghapc} and computes

$$a_{bngha} = \sum_{p,c} \bar{z}_{bnhpc} A_{ghapc} + \beta_{gha}, \tag{6}$$

where $a \in \{1, \dots, 6\}$ indexes half-turn axes and $\bar{z}$ denotes the bias-corrected tokenizer output. Appending a learned null score and applying softmax produces pose weights.

Every one of the first 11 Transformer blocks and its three heads retains an independent 4×12 field. A single pose probe may be reused by G consecutive blocks, requiring $\lceil 11/G \rceil$ groups. This separates the cost of recognizing token pose from the layer-specific choice of where to look. Feature Look is omitted from the final block because patch-to-patch bias there cannot affect that same block's CLS output. If Image and Feature Look are enabled together, their bases can be concatenated because the structured additive attention construction is linear.

3.7 Transformer backbone

All ViT variants use a DeiT-Tiny-sized backbone with embedding dimension 192, 12 blocks, three attention heads, MLP ratio 4, and a CLS classifier. Absolute positional embeddings are either enabled or disabled as an explicit ablation. The large compact matching operations use mixed precision, while the small routing softmax is evaluated in FP32.

4 Experiments

4.1 Dataset and training protocol

We use ImageNet-100 with 100 classes, 130,000 training images, and 5,000 validation images. All main-table models are trained from scratch for 20 epochs at 224×224 resolution with no pretrained weights or teacher. Training uses two NVIDIA T4 GPUs, batch size 256 per GPU (global batch 512), AdamW, initial learning rate 5×10^{-4}, minimum learning rate 10^{-6}, weight decay 0.05, two warmup epochs, cosine decay, label smoothing 0.1, gradient clipping at 1.0, mixed precision, and seed 0.

The training transform comprises random resized crop with bicubic interpolation, horizontal flip, RandAugment with two operations and magnitude 9, ImageNet normalization, and random erasing with probability 0.25. Validation resizes to 256, center-crops to 224, and applies ImageNet normalization. We report the best validation top-1 and the top-5 from the same epoch. Only one seed is complete, so the study is a controlled architectural ablation rather than a statistically conclusive benchmark.

4.2 Compared configurations

Standard is the ordinary square 16×16 DeiT patch projection. **Hex only** changes patch centers and support without rotating prototypes. **Hex** $\mathcal{G}_{4,4}$ and **Hex** $\mathcal{G}_{6,3}$ use the two allocations defined in Sec. 3.3: in $\mathcal{G}_{D,q}$, D counts tested half-turn poses and q gives the per-radius base sampling factor. For each rotating tokenizer, we compare PE only, Image Look only, and PE+Image Look. We then fix Hex $\mathcal{G}_{6,3}$ and PE and vary the Feature Look sharing span G.

The main comparison additionally includes three rotation-aware controls. **Equi-ViT/GMR** adapts the published Gaussian-mixture-ring operator [Chen et al., 2026] to the common DeiT-sized classifier, **ARC Adaptive** predicts bounded input-conditioned rotations for a bank of patch kernels [Pu et al., 2023], and **GE-ViT p4 local** preserves a four-state orientation axis through a local group-attention backbone [Xu et al., 2023]. These are operator-level adaptations under the shared ImageNet-100 recipe, not claims to reproduce the original papers' full datasets and training systems.

Parameter counts come from stored model summaries. Cross-version wall times are omitted because shared Kaggle sessions do not support controlled hardware comparisons.

5 Results

5.1 Main comparison and local rotation response

Table 1: Local re-evaluation of the six saved checkpoints. The 0° columns evaluate the unrotated 5,000-image ImageNet-100 validation set. Local means average the 12 nonzero rotations from −30° to +30° in 5° increments. JSD is the mean Jensen–Shannon divergence from each model's own 0° predictive distribution and is confidence-sensitive.

	0° validation		Local rotations			
Model	Top-1	Top-5	Mean Top-1	Mean Top-5	100× Mean JSD ↓	Params (M)
Standard DeiT-Tiny	51.54	79.12	49.91	77.45	5.27	5.544
SHARE $\mathcal{G}_{4,4}$ + Image Look	53.68	80.46	51.25	78.89	5.65	5.501
SHARE $\mathcal{G}_{6,3}$ + dual Look	55.52	81.80	52.82	**79.98**	5.64	5.497
Equi-ViT / GMR	41.52	70.40	41.08	70.12	**3.22**	5.424
ARC Adaptive	49.38	76.56	47.76	75.70	5.25	5.986
GE-ViT p4 local	**56.06**	**82.04**	**52.94**	79.45	5.88	5.509

For predictive distributions $p_{i,\theta}$ and $p_{i,0}$, let $m_{i,\theta} = \frac{1}{2}(p_{i,\theta} + p_{i,0})$. We report

$$\overline{\mathrm{JSD}}_{100} = \frac{100}{12N} \sum_{\theta \in \{\pm 5^\circ, \ldots, \pm 30^\circ\}} \sum_{i=1}^{N} \left[\tfrac{1}{2} D_{\mathrm{KL}}(p_{i,\theta} \| m_{i,\theta}) + \tfrac{1}{2} D_{\mathrm{KL}}(p_{i,0} \| m_{i,\theta})\right]. \tag{7}$$

Bold, underlined, and blue values denote the best, second-best, and third-best result in each evaluation column, respectively.

Table 1 separates ordinary validation accuracy from local rotation behavior. SHARE $\mathcal{G}_{6,3}$ is within 0.12 top-1 points of GE-ViT after averaging the local rotations and has the highest mean top-5, while retaining fewer parameters than Standard. Relative to Standard, its local means improve by 2.91 top-1 and 2.53 top-5 points. GE-ViT has the strongest unrotated and mean local top-1 results, consistent with retaining an explicit orientation axis throughout its backbone. ARC Adaptive is below Standard under this short from-scratch recipe. Equi-ViT/GMR has the smallest JSD but also much lower accuracy; its small probability drift must therefore not be read as superior classification robustness in isolation.

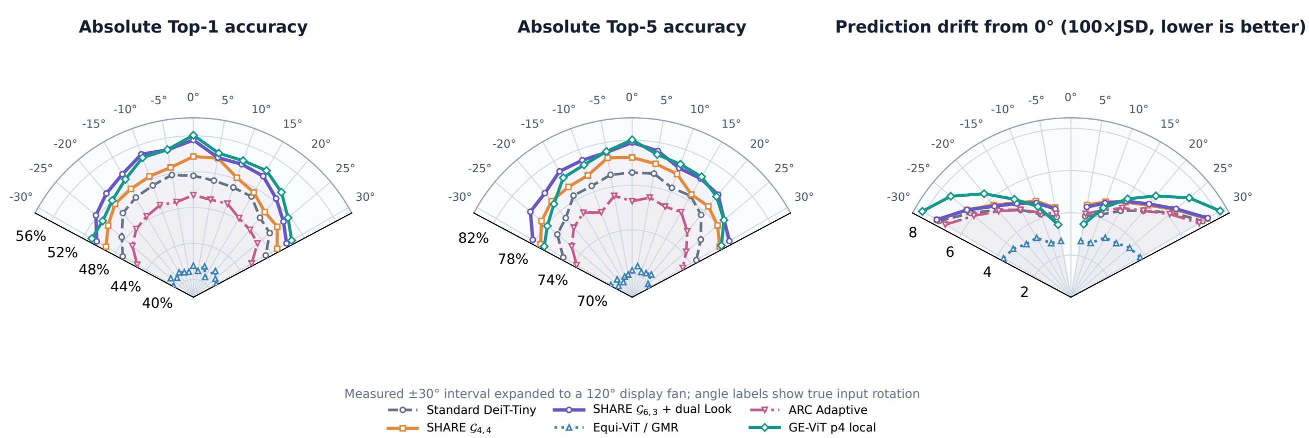

Figure 3: Local rotation response on all 5,000 ImageNet-100 validation images. Left and middle show absolute top-1 and top-5 accuracy. Right shows 100× Jensen–Shannon drift from each model's own 0° distribution; the tautological zero-angle JSD point is omitted, leaving the positive and negative branches visually separate. The measured ±30° range is expanded to a 120° display fan while labels retain the true input angles. Every model receives identical bilinearly rotated inputs.

Figure 3 exposes three complementary effects. First, top-1 declines smoothly with increasing absolute rotation for all models, rather than failing at one isolated angle. Second, the top-5 panel shows that SHARE $\mathcal{G}_{6,3}$ retains a stronger candidate-class set across the local interval even where GE-ViT has slightly higher top-1. Third, JSD measures probability movement that top-k accuracy cannot expose. Equi-ViT/GMR's low JSD accompanies low confidence and low accuracy, whereas GE-ViT's larger JSD coexists with the strongest top-1; the three panels must therefore be read together. SHARE $\mathcal{G}_{6,3}$ also shows a small sign asymmetry: positive rotations average 0.77 top-1 points below matched negative rotations. Because Standard, ARC, Equi-ViT/GMR, and GE-ViT do not share this sign, we treat it as a model-specific diagnostic of the half-angle/Look pathway, not as evidence of a universal preprocessing bias or exact equivariance.

Table 2: Tokenizer geometry and positional-signal ablation under the aligned 20-epoch ImageNet-100 recipe. Geometry symbols follow Sec. 3.3. Δ is the percentage-point change from Standard. Top-5 is taken at the best-top-1 epoch.

Tokenizer	PE	Image Look	Top-1	ΔTop-1	Top-5	ΔTop-5	Params (M)
Standard	✓	–	51.52	–	79.12	–	5.544
Hex only	✓	–	51.30	−0.22	79.14	+0.02	5.597
Hex $\mathcal{G}_{4,4}$	✓	–	53.56	+2.04	80.40	+1.28	5.486
Hex $\mathcal{G}_{4,4}$	–	✓	50.26	−1.26	77.72	−1.40	5.463
Hex $\mathcal{G}_{4,4}$	✓	✓	53.70	+2.18	80.46	+1.34	5.501
Hex $\mathcal{G}_{6,3}$	✓	–	54.54	+3.02	80.46	+1.34	5.464
Hex $\mathcal{G}_{6,3}$	–	✓	52.42	+0.90	79.48	+0.36	5.453
Hex $\mathcal{G}_{6,3}$	✓	✓	**55.04**	+3.52	**81.80**	+2.68	5.491

5.2 Tokenizer and Image Look ablation

Directions versus ring resolution. The $\mathcal{G}_{4,4}$ configuration evaluates four poses over a half turn and uses $q = 4$; $\mathcal{G}_{6,3}$ evaluates six poses while using $q = 3$. Thus pose count and polar storage density move in opposite directions in this experiment. They are a coupled geometry choice rather than an isolated direction-count ablation.

Table 2 shows that the plain Hex tokenizer does not improve top-1, indicating that a staggered lattice alone is insufficient. Rotating prototype matching is the main source of gain: PE-only $\mathcal{G}_{4,4}$ and $\mathcal{G}_{6,3}$ improve over Standard by 2.04 and 3.02 points, respectively. $\mathcal{G}_{6,3}$ outperforms $\mathcal{G}_{4,4}$ under all three positional configurations. Because this comparison jointly changes the number of tested directions and samples stored per ring, it supports the complete six-direction allocation, not a claim about either factor in isolation.

Image Look does not replace absolute location: Look-only trails PE+Look in both geometry settings. It is nevertheless complementary to PE. For $\mathcal{G}_{6,3}$, adding Image Look increases top-1 from 54.54 to 55.04 and top-5 from 80.46 to 81.80. The best model exceeds Standard by 3.52 top-1 and 2.68 top-5 points while using 5.491M instead of 5.544M parameters. The gain is therefore not explained by a larger parameter budget.

5.3 Feature Look sharing span

Figure 4 reveals a non-monotonic sharing trade-off. Recomputing a distinct pose detector for every block ($G = 1$) is not best. Moderate sharing with $G = 3$ or 4 improves top-1, and $G = 4$ reaches 55.18%, slightly above the 55.04% Image Look model. More aggressive sharing ($G = 6$ or 12) falls back to approximately 54.5%. This is consistent with a pose cue that is reusable across nearby layers but too coarse when globalized across the network. Since all rows use one seed and differences are small, this remains a bounded interpretation rather than a statistical conclusion.

The optimum changes when Image Look is also active. As shown in Fig. 4, the dual-Look model peaks at $G = 3$ with 55.54% top-1, 0.50 points above PE+Image Look and 0.54 points above the PE+Feature Look model at the same span. The older dual-Look $G = 4$ point used the pre-optimization expanded kernel, so it is retained for provenance rather than treated as a controlled kernel-speed comparison. Together, the curves support an interaction between branch placement and sharing span rather than unconditional additivity or redundancy.

The lower panel reports the architectural number of pose-probe evaluations; wall-clock time is omitted because cross-version timing was not controlled. Each block still retains its own 4×12 output field.

5.4 Learned directed fields

Figure 5 shows that the learned fields are neither uniform across heads nor constant across depth. This visualization verifies variation in the implemented directed fields, but it does not establish semantic part discovery, causal explanation, or exact rotation behavior.

6 Discussion

6.1 What the ablations establish

Within the aligned setup, three conclusions are supported. First, changing the sampling lattice alone is insufficient; learned multi-pose prototype matching is the important tokenizer change. Second, absolute PE and Image Look are complementary: the former supplies persistent token identity, while the latter adds image-conditioned directed relations. Third, Feature Look

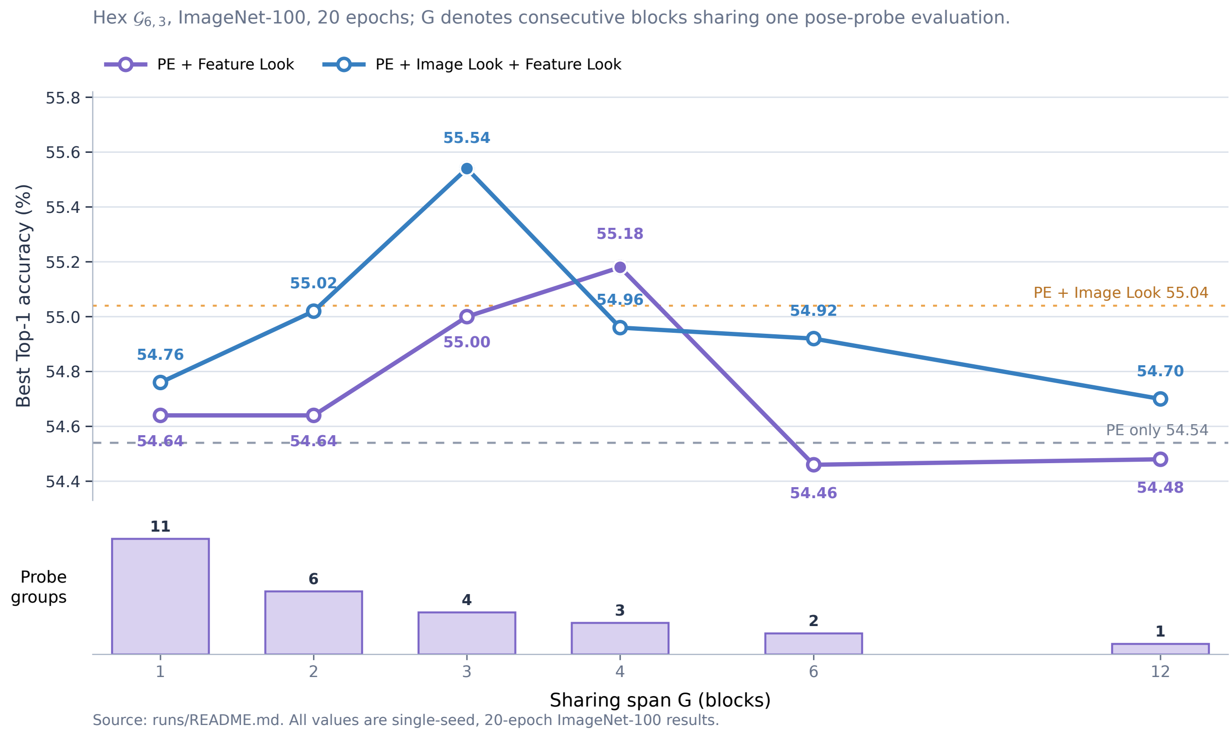

Figure 4: Feature Look probe-sharing ablation. Purple uses PE+Feature Look; blue additionally enables Image Look. The lower panel reports the architectural number of probe evaluations. All values are single-seed 20-epoch results.

can reuse one pose detector over a short layer span without sharing the layer-specific attention field, but global probe sharing loses accuracy. Its best span also depends on placement: G4 is best with PE alone, whereas G3 is best when Image Look is active, where the two directed branches provide the strongest observed result.

The six-direction allocation is consistently better than the four-direction allocation in the tested configurations. It also uses a more compact angular storage per radius, so the present experiment cannot attribute the gain only to direction count. This coupling is an explicit target for a future factorial ablation.

6.2 Limitations

The current evidence has five important limits. (1) Only one random seed is complete, so differences of a few tenths of a point may reflect training variance. (2) The 20-epoch schedule is short and may rank architectures differently from a converged recipe. (3) The method samples a finite set of rotations; interpolation, image boundaries, and the unconstrained backbone prevent a claim of exact equivariance. (4) Evaluation covers one classification dataset and does not establish ImageNet-1K, detection, segmentation, robustness, or transfer gains. (5) shared Kaggle wall times are insufficient for a hardware-efficiency claim.

The visualizations expose how the implemented routing behaves, but the learned fields are operational model components rather than causal explanations of a prediction. Their semantic interpretation must remain descriptive.

6.3 Next validation

The highest-priority experiment is an angle–scale controlled benchmark that trains on one set of poses and evaluates on interleaved unseen poses. This would directly test whether shared prototypes improve interpolation. A factorial tokenizer experiment should independently vary tested direction count and ring storage density. Multi-seed ImageNet-100 runs under a longer schedule are needed for confidence intervals. Longer ImageNet-100 training and ImageNet-1K evaluation would then test whether the observed gains persist beyond the current short, single-dataset setting.

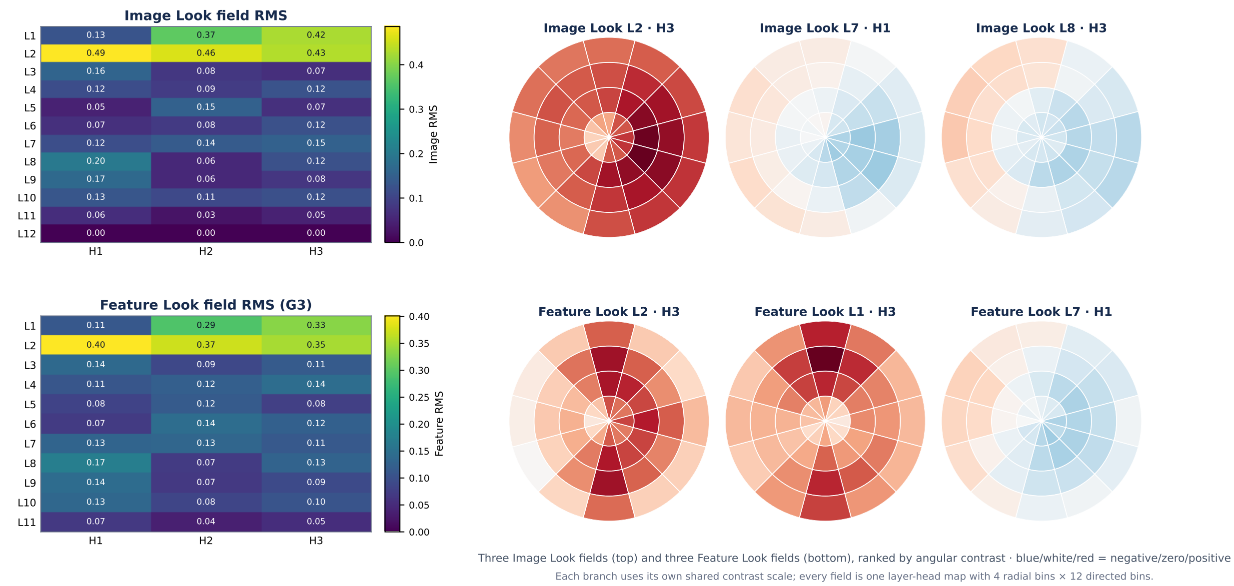

Figure 5: Learned parameters from the best dual-Look model ($\mathcal{G}_{6,3}$ + PE + Image Look + G3 Feature Look). Left: per-layer, per-head RMS magnitude for Image and Feature Look, shown with separate color scales. Right: the three layer–head fields with the largest angular contrast from each branch, measured after removing the mean of every radial ring. Each signed 4×12 map is shown separately rather than mixing heads as RGB. G3 reuses one pose probe for each consecutive three-layer group, while every active layer retains its own output field. Nonuniform fields are an implementation-level observation, not a causal interpretability claim.

7 Conclusion

We presented SHARE-ViT, a geometric interface between images and global self-attention. Hex-centered multi-scale patches are matched against rotating variable-resolution polar prototypes and compressed through null-softmax circular moments. Image Look converts independent raw-image evidence into directed attention fields, while Feature Look obtains a lighter pose probe directly from token coordinates and exposes a useful local layer-sharing scale. In the current aligned ImageNet-100 study, the six-direction Image Look model improves top-1 from 51.52% to 55.04%; Feature Look with PE reaches 55.18%; and their best observed combination reaches 55.54% at G3, without increasing the baseline parameter count. The results motivate explicit local pose and scale as complements to positional encoding, while leaving exact equivariance, statistical confirmation, and broader task generalization open.

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